8-Queen Problem

AIM: To write a Python program to solve the 8-Queen Problem by placing eight queens on a chessboard so that no two queens attack each other.

Algorithm:

1. Start the program.
2. Initialize an empty 8 × 8 chessboard.
3. Place a queen in the first row.
4. Check whether the selected position is safe from other queens.
5. If the position is safe, place the queen and move to the next row.
6. If no safe position exists, backtrack to the previous row and change the queen's position.
7. Repeat until all eight queens are placed safely.
8. Display the solution.
9. Stop the program.

Code :

```python
BOARD_SIZE = 8

def is_safe(board, row, col):
    for i in range(col):
        if board[row][i] == 1:
            return False

    i, j = row, col
    while i >= 0 and j >= 0:
        if board[i][j] == 1:
            return False
        i -= 1
        j -= 1

    i, j = row, col
    while i < BOARD_SIZE and j >= 0:
        if board[i][j] == 1:
            return False
        i += 1
        j -= 1

    return True

def solve_n_queens_util(board, col, solutions):
    if col >= BOARD_SIZE:
        solutions.append([row[:] for row in board])
        return

    for row in range(BOARD_SIZE):
        if is_safe(board, row, col):
            board[row][col] = 1
            solve_n_queens_util(board, col + 1, solutions)
            board[row][col] = 0

def print_board(board):
    for r in range(BOARD_SIZE):
        print(" ".join("Q" if board[r][c] == 1 else "." for c in range(BOARD_SIZE)))
```

Output :

```
PS D:\CSA1702-AI\Experiments> & C:\python311\python.exe d:/CSA1702-AI/Experiments/exp02_8_queens.py
Total solutions: 92

Solution 1:
Q . . . . . . .
. . . . . . Q .
. . . . Q . . .
. . . . . . . Q
. Q . . . . . .
. . . Q . . . .
. . . . . Q . .
. . Q . . . . .

Solution 2:
Q . . . . . . .
. . . . . . Q .
. . . Q . . . .
. . . . . Q . .
. . . . . . . Q
. Q . . . . . .
. . . . Q . . .
. . Q . . . . .

Solution 3:
Q . . . . . . .
. . . . . Q . .
. . . . . . . Q
. . Q . . . . .
. . . . . . Q .
. . . Q . . . .
. Q . . . . . .
. . . . Q . . .
```

RESULT: Thus, the Python program to solve the 8-Queen Problem was executed successfully.